Level 1: Isogyric

Co-reactants



[H][H]

Candidate products



[*]

Level 2: Isodesmic (candidates built directly from the bond types of the target)

Co-reactants



[*]



[H][H]

Candidate products



[*][*]



[*]=[*]



[*]#[*]

Level 3: Hypohomodesmotic

Co-reactants



[*][*]



[*]=[*]



[*]#[*]

Candidate products



[*][*][*]



[*]=[*][*]



[*]#[*][*]



[*][*](*)[*]



[*][*](=[*])[*]



[*][*@]([*])([*])[*]



[*]=[*]=[*]

X: any atom; Y: any non-carbon atom; hydrogens are added to complete the valence.

Isodesmic: X stands for the elements of the target; one X–X', X=X' or X≡X' species per bond type present, the hydride of each element, and H_2.